ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** celestial bodies	Bodies beyond the Earth.
2	**(d)** Mercury	Smallest planet.
3	**(a)** new-moon day	Amavasya.
4	**(b)** star	It produces its own light and heat.
5	**(b)** Jupiter	Jupiter.
6	**(b)** artificial satellites	They are launched by rockets.
7	**(c)** their light passes through the moving air of the atmosphere	Atmospheric disturbance.
8	**(c)** Mars	Iron oxide gives red colour.
9	**(c)** Aryabhata	Launched in 1975.
10	**(c)** Moon	It reflects sunlight.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Natural object in space (1); Sun and Moon (1).	1+1
12	Mercury, Venus, Earth, Mars (1); Jupiter, Saturn, Uranus, Neptune (1).	1+1
13	Communication (1); Weather forecast (1).	1+1
14	Starlight passes through moving air (1); The atmosphere disturbs the light (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Moon has no light of its own (1); It reflects sunlight (1); It revolves around Earth (1).	1+1+1
16	Shows north (1); Remains nearly fixed (1); Helps to find direction at night (1).	1+1+1
17	The Sun's light is very bright (1); Stars are still there (1); Their light is faint in comparison (1).	1+1+1
	OR: Light and heat (1); Plants make food (1); Drives weather and water cycle (1).	1+1+1
18	Thick atmosphere (1); Traps heat (1); Mercury has almost no atmosphere (1).	1+1+1
	OR: Patterns are recognised (1); Stars named in groups (1); Used for navigation (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sun at the centre (1); Mercury: nearest to the Sun (1); Jupiter: largest (1); Saturn: rings (1); Mars: red (1).	1+1+1+1+1
	OR: Planets orbit the Sun (1); Moon orbits Earth (1); Orbit is a path (1); Held by gravity (1); Revolution takes different times (1).	1+1+1+1+1
20	Moon reflects sunlight (1); Half of it is always lit (1); Different parts visible as it orbits (1); New moon: not visible (1); Full moon: whole lit face seen (1).	1+1+1+1+1
	OR: Stars are very far (1); They look like points of light (1); Air disturbs the light (1); Planets are closer (1); Appear as small discs (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that the Moon makes its own light; it only **reflects** sunlight.
- Writing nine planets or the wrong order; there are eight, starting with Mercury nearest to the Sun.
- Thinking the Moon itself changes shape; we only see different amounts of its lit half as it goes around the Earth.
- Mixing Chandrayaan (Moon) with Mangalyaan (Mars); Aryabhata was a satellite.
- Saying the Pole star is the brightest star or moves around; it appears nearly fixed and shows **north**.
- Confusing a star with a planet: the Sun is a star and the Earth is a planet.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Saturn	Saturn's rings.
2	**(b)** Indian Space Research Organisation	It is India's space agency.
3	**(a)** constellation	Such as Ursa Major.
4	**(c)** Venus	Thick atmosphere traps heat.
5	**(b)** Moon	Lunar missions.
6	**(b)** Saptarishi	Seven sages.
7	**(b)** Earth	Third from the Sun.
8	**(d)** Mars	Mars Orbiter Mission.
9	**(b)** Pole star	It lies near the Earth's axis direction.
10	**(a)** Moon	It revolves around Earth.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water covers most of it (1); It looks blue from space (1).	1+1
12	Small rocky body orbiting the Sun (1); Found between Mars and Jupiter (1).	1+1
13	Group of stars forming a pattern (1); Ursa Major or Orion (1).	1+1
14	Different shapes of the lit part (1); Seen from Earth during a month (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Star has its own light (1); Planet reflects light (1); Planets revolve around a star (1).	1+1+1
16	Water (1); Air with oxygen (1); Suitable temperature (1).	1+1+1
17	No oxygen (1); Extreme temperatures (1); No liquid water (1).	1+1+1
	OR: Mercury is the nearest (1); Neptune is the farthest (1); Order from Mercury to Neptune (1).	1+1+1
18	Planet is large (1); Asteroid is small and rocky (1); Both orbit the Sun (1).	1+1+1
	OR: Ice and dust (1); Tail grows near Sun (1); Moves in a long orbit (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Saptarishi is a group of seven stars (1); The two stars of the bowl point to the Pole star (1); Pole star lies in the north (1); Facing it, north is ahead (1); Other directions follow (1).	1+1+1+1+1
	OR: Asteroid: rocky (1); Comet: ice and dust with tail (1); Meteor: streak of light (1); Meteorite: reaches ground (1); Smaller than planets (1).	1+1+1+1+1
20	Communication (1); Weather (1); Navigation (1); Aryabhata, INSAT (1); Chandrayaan, Mangalyaan (1).	1+1+1+1+1
	OR: Craters (1); No water seen (1); No air (1); Big temperature changes (1); Footprints stay (1).	1+1+1+1+1

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- Writing nine planets or the wrong order; there are eight, starting with Mercury nearest to the Sun.
- Thinking the Moon itself changes shape; we only see different amounts of its lit half as it goes around the Earth.
- Mixing Chandrayaan (Moon) with Mangalyaan (Mars); Aryabhata was a satellite.
- Saying the Pole star is the brightest star or moves around; it appears nearly fixed and shows **north**.
- Confusing a star with a planet: the Sun is a star and the Earth is a planet.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Sun	Solar mission.
2	**(a)** north direction	It shows north.
3	**(d)** 27 days	About a month.
4	**(b)** communication and weather forecasting	Many uses.
5	**(b)** Sun	They orbit the Sun.
6	**(a)** phases of the Moon	Waxing and waning.
7	**(c)** asteroids	Mostly between Mars and Jupiter.
8	**(d)** eight	Mercury to Neptune.
9	**(d)** full-moon day	Poornima.
10	**(d)** ice, dust and rock	It develops a tail near the Sun.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Different amount of its lit half is seen (1); Its position changes around Earth (1).	1+1
12	Join the two end stars of the bowl (1); Extend the line to find it (1).	1+1
13	Man-made object orbiting the Earth (1); Aryabhata or INSAT (1).	1+1
14	Chandrayaan (1); Mangalyaan (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Takes about 27 days (1); Moon orbits Earth (1); Earth and Moon orbit the Sun (1).	1+1+1
16	Takes pictures of clouds (1); Sends data (1); Helps predict rain and storms (1).	1+1+1
17	Sun and all bodies around it (1); Planets, moons, asteroids, comets (1); Held by Sun's gravity (1).	1+1+1
	OR: New moon (1); Half moon and crescent (1); Full moon (1).	1+1+1
18	ISRO (1); Aryabhata satellite (1); Chandrayaan and Mangalyaan (1).	1+1+1
	OR: It rotates once in the time it revolves (1); So same side faces Earth (1); Far side is not seen (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Star of our system (1); Source of heat and light (1); Photosynthesis (1); Water cycle (1); Winds and seasons (1).	1+1+1+1+1
	OR: Place a lamp as the Sun (1); Hold a ball as the Moon (1); Move it around (1); Lit part changes (1); Shows phases (1).	1+1+1+1+1
20	Earth: water, oxygen (1); Mars: cold, thin air (1); Venus: very hot (1); Earth is the only known with life (1); Mars is being studied (1).	1+1+1+1+1
	OR: About the origin of the universe (1); About other planets (1); New technology (1); Weather and communication (1); Protect Earth (1).	1+1+1+1+1

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